Federated Learning Across Hospitals with Differential Privacy: Literature Review, Research Directions, and a Plan for Orlando (Fall 2026)

The team should pick the privacy–fairness audit (the old Idea D), but change one part of it: on a 4-hospital dataset such as Fed-Heart-Disease, use patient-level (sample-level) differential privacy, not hospital-level (client-level) differential privacy, and first build an evaluation harness that reports per-hospital, worst-hospital, local/federated/pooled, and bytes-sent results. As of October 2026, our search found no published paper that measures subgroup fairness (sex, age, race) of a federated healthcare model across several privacy budgets, so the gap named in the earlier brief is still open.

TL;DR

- The core gap is confirmed. The largest review of differential privacy (DP) in medical deep learning (Mohammadi et al., npj Digital Medicine, 74 studies through March 2025) found that only five studies audit fairness (Chin-Cheong et al., Kaess et al., and three by Tayebi Arasteh et al.), that most report fairness at only one privacy level, and that the multi-ε fairness studies it names (all three by Tayebi Arasteh et al.) are centralized (not federated). A targeted search of January 2025–October 2026 work found no federated, multi-ε, demographic-subgroup audit on medical data.
- Most other claims in the earlier brief check out, with corrections: FLamby (NeurIPS 2022) does show that only the two tabular datasets reach pooled performance, but it also shows that for three datasets federated training gave no benefit over local training; the Medical Image Analysis 2025 review numbers (19/107, 41/107, 10/107, 84%) are correct; the chest X-ray numbers (89.7 → 87.4 at $\varepsilon = 7.9$ → 84.0 at $\varepsilon = 0.5$) are correct but come from a centralized study; MedHE (42× compression, 54% overhead) is an unreviewed single-author preprint and should be treated with caution.
- Orlando should, in this order: (1) add per-hospital test evaluation, local/federated/pooled comparison, worst-hospital reporting, and a bytes counter to his FedAvg harness; (2) add patient-level DP-SGD with Opacus and a Rényi DP accountant, sweeping $\varepsilon \in \{0.5, 1, 3, 8, \infty\}$; (3) add sex and age subgroup metrics with bootstrap confidence intervals; (4) start PhysioNet credentialing for eICU now, because scaling to eICU's 208 hospitals is what will make the results publishable.

Part 0. Key terms

- **Federated learning (FL):** training one model across several data holders (here, hospitals) without moving raw data. Each hospital trains locally and sends only model parameters to a server, which combines them.

- **Client:** one data holder, here one hospital. **Cross-silo FL** means few reliable clients with medium or large datasets (FLamby: 2–50 clients). neurips arxiv
- **FedAvg (Federated Averaging):** the basic FL algorithm (McMahan et al., AISTATS 2017). Each round the server sends the model to clients, each client runs a few steps of stochastic gradient descent on its own data and sends the result back, and the server averages the models, weighting by number of training samples. arxiv
- **Local training:** each hospital trains only on its own data. **Pooled (centralized) training:** all data in one place, one model. Pooled is the upper reference point, local the lower; FL should land between or match pooled.
- **Non-IID data:** data not identically distributed across clients (different patient mixes, scanners, label rates). This is the main reason FL falls short of pooled training. arxiv
- **Differential privacy (DP):** a guarantee that training output changes very little whether or not any one individual's data was included. **ε (epsilon)** is the privacy budget: smaller means stronger privacy and usually lower accuracy ($\varepsilon \approx 1$ is strict, $\varepsilon \approx 8$–10 moderate). **δ (delta)** is a small probability the guarantee fails; common practice is $\delta \leqslant 1/n$ for dataset size n. nature nih
- **DP-SGD:** standard DP training (Abadi et al., CCS 2016). Compute each example's gradient, **clip** it to length at most C (the **clipping norm**), sum, and add Gaussian noise of size **noise multiplier** σ times C. nature nih
- **Privacy accountant:** software that tracks privacy spent over many steps and outputs a final (ε, δ). The **Rényi DP (RDP) accountant** gives tighter values than simply adding per-step ε. nih
- **Sample-level vs. client-level DP:** sample-level DP hides whether one patient record was used; client-level DP hides whether a whole hospital took part. Client-level is much stronger and costlier and works well only with many clients.
- **Local vs. central DP:** in local DP each client adds noise before sending; in central DP noise is added at aggregation, which requires trusting the server or using secure aggregation. Central DP gives better accuracy at the same ε. nih flower
- **Secure aggregation:** a protocol that lets the server learn only the sum of updates. **Homomorphic encryption (HE):** encryption that allows arithmetic on encrypted numbers. arxiv
- **AUC (AUROC):** 0.5 is random, 1.0 is perfect ranking. **Balanced accuracy:** average per-class recall. **Dice score:** overlap of predicted and true segmentation masks (0 to 1). **C-index:** fraction of patient pairs whose risks are correctly ordered.
- **Statistical parity difference (PtD):** difference in positive-prediction rate between two groups; the most common fairness metric in DP-medical work (Mohammadi et al.). nih
- **Gradient inversion attack:** reconstructing training data from a model update. **Membership inference attack (MIA):** guessing whether a given patient was in the training set.

Part 1. Literature review

1.1 Benchmarks and reviews that define the field

FLamby: Datasets and Benchmarks for Cross-Silo Federated Learning in Realistic Healthcare Settings. Ogier du Terrail et al. (Owkin, Inria, EPFL, and others), NeurIPS 2022 Datasets and Benchmarks track (arXiv 2210.04620). Seven healthcare datasets, each split by real origin (hospital, hospital group, or scanner maker), with fixed per-client train/test splits, a baseline model, and a metric (Table 1). arxiv It compares FedAvg, FedProx, SCAFFOLD, Cyclic, FedAdam, FedYogi, and FedAdagrad against local and pooled training, with 100 local updates per round and 5 runs (1 for Fed-LIDC-IDRI); results are averaged uniformly over clients. arxiv arxiv

Headline results (Section 4, Figure 2):

- "No local training or FL strategy is able to reach a performance on par with the pooled training, except for Fed-TCGA-BRCA and Fed-Heart-Disease," both tabular with linear models. arxiv arxiv
- For Fed-KITS2019 and Fed-ISIC2019, some FL strategies beat local training but stay below pooled. arxiv arxiv
- For Fed-Camelyon16, Fed-LIDC-IDRI, and Fed-IXI, "the current results do not indicate any benefit in collaborative training"; on Fed-LIDC-IDRI several strategies failed to converge (Dice of zero). arxiv arxiv
- Where FL helps, FedOpt variants (FedAdam, FedYogi, FedAdagrad) are best; Cyclic is worst; FedAvg is competitive. arxiv arxiv
- Results appear only as bar charts; exact numbers must come from the FLamby GitHub benchmark outputs.
- The conclusion says the benchmark "does not integrate all constraints of cross-silo FL, especially privacy aspects"; arxiv Appendix L.1 has only a small DP-FedAvg (Opacus) example on Fed-Heart-Disease, with accuracy falling as ε approaches 0. arxiv

What this means: Fed-Heart-Disease is good for debugging a harness because FL already matches pooled there, but poor for showing a new FL algorithm helps. It is still useful for privacy and fairness questions, because DP noise will open a gap.

Correction to the earlier brief: add that three of seven datasets showed no benefit from collaboration. FLamby lists Fed-Heart-Disease with 13 input features after preprocessing, not 16; arxiv if Orlando's harness reports 16, he should check his preprocessing. arxiv

From Challenges and Pitfalls to Recommendations and Opportunities: Implementing Federated Learning in Healthcare. Ming Li, Pengcheng Xu, Junjie Hu, Zeyu Tang, Guang Yang (Imperial College London and others), Medical Image Analysis, 2025 (arXiv 2409.09727).

PRISMA review of 107 healthcare FL studies up to May 2024, from 1,149 records. arxiv Numbers from Table 1 and Sections 4–5, matching the earlier brief:

- 10 of 107 reported real-world deployment; 97 were simulations. sciencedirect arxiv
- 19 of 107 studied communication efficiency; 41 of 107 protected model updates. arxiv
- 84% did not define per-client train/test splits; only 17 gave per-client details. arxiv
- 37 used natural splits; 67 used artificial splits. arxiv
- 26 of 107 addressed FL-specific issues; only 3 discussed fairness or incentives; only 5 stated they used pretrained or foundation models. arxiv
- 78 used custom FL code; 13 used open-source frameworks (Flower, NVIDIA FLARE, SubstraFL, TFF, OpenFL, PySyft, Fed-BioMed). arxiv
- It recommends a dedicated test set inside each client. arxiv

Differential privacy for medical deep learning: methods, tradeoffs, and deployment implications. Marziyeh Mohammadi, Mohsen Vejdanihemmat, Mahshad Lotfinia, Mirabela Rusu, Daniel Truhn, Andreas Maier, Soroosh Tayebi Arasteh (FAU Erlangen, RWTH Aachen, Stanford). npj Digital Medicine 9:93 (published 3 January 2026; preprint arXiv 2506.00660). nature nih The earlier brief called this "2025"; the journal version is dated 2026. Scoping review of 74 studies through March 2025; 67 used DP-SGD. nih nih

Key findings:

- DP-SGD "can maintain clinically acceptable performance under moderate privacy budgets ($\varepsilon \approx 10$), particularly in imaging," but "strict privacy ($\varepsilon \approx 1$) often leads to substantial accuracy loss, with amplified degradation in smaller or heterogeneous datasets." nih +2
- Gaussian mechanism plus RDP accountant dominates, usually via Opacus or TensorFlow Privacy; federated studies often omit DP parameters. nih
- Pretraining is "a consistent and effective strategy"; DP-SGD from scratch "failed to learn meaningful features." Batch normalization is avoided; group normalization is the usual replacement. nih
- Fairness: only five studies examine fairness under DP (Table 4); "most studies report fairness at only a single privacy level"; the review names three works (all by Tayebi Arasteh et al.) as examples of studies that "systematically analyze fairness across multiple ε values," using the wording "such as," so the list may not be complete; "fairness-aware techniques … are almost entirely absent."

A companion npj Digital Medicine editorial (2026) restates that "only 5/74 studies audited the fairness of their differential privacy techniques" and that none examined intersectional groups. nature This confirms the earlier brief's "about five" claim.

Federated Learning for Medical Image Classification: A Comprehensive Benchmark. Zhekai Zhou, Guibo Luo, Mingzhi Chen, Zhenyu Weng, Yuesheng Zhu, arXiv 2504.05238 (April 2025).

Compares many FL algorithms, including FedBN and a one-shot method, on several medical imaging datasets, plus communication and compute cost. Abstract: "No single algorithm consistently delivers optimal performance across all medical federated learning scenarios, and many optimization algorithms may underperform." arxiv researchgate This confirms the earlier brief.

1.2 Large real-world hospital FL studies

EXAM: Federated learning for predicting clinical outcomes in patients with COVID-19. Ittai Dayan, Holger Roth, Aoxiao Zhong, et al., Nature Medicine 27:1735–1743, October 2021. harvard 20 institutes trained one model to predict 24- and 72-hour oxygen needs of COVID-19 emergency patients from 20 inputs (19 medical-record features plus a chest X-ray), nature using a ResNet-34 image branch and a Deep & Cross network. frontiersin Average AUC was above 0.92 for both horizons. harvard Average AUC across sites rose from 0.795 (local) to 0.920 (federated), a 16% gain; generalizability (AUC on other sites' test data) rose from 0.667 to 0.920, a 38% gain. harvard It was validated at three independent sites. globenewswire No formal DP with a reported ε was used.

Towards fair decentralized benchmarking of healthcare AI algorithms with the Federated Tumor Segmentation (FeTS) challenge. Maximilian Zenk, Ujjwal Baid, Sarthak Pati, Spyridon Bakas, and others, Nature Communications 16:6274, 8 July 2025. nature +2 Glioma segmentation from multi-parametric MRI, scored with Dice. Task 1 compared aggregation and client selection in simulated FL on BraTS data (the 2024 edition used 1,473 cases with institution labels, per FedPOD, arXiv 2512.20610); arxiv Task 2 evaluated algorithms on data at 32 sites. Adaptive aggregation helped and client sampling saved compute, but "while average generalization was good, worst-case performance highlighted data-specific failures" (Luxembourg Institute of Health summary). lih This confirms the earlier brief.

Other deployments. The Medical Image Analysis review lists CODA (8 Canadian hospitals, COVID-19), MELLODDY (10 pharmaceutical companies), HealthChain and DRAGON (European hospitals), and Owkin's triple-negative breast cancer work. arxiv These are mostly reported without formal DP accounting.

1.3 DP in medical ML: key primary studies

Preserving fairness and diagnostic accuracy in private large-scale AI models for medical imaging. Soroosh Tayebi Arasteh, Alexander Ziller, et al., Communications Medicine 4:46, 14 March 2024 (arXiv 2302.01622). nature +2 Centralized DP-SGD. ResNet-9 with group normalization, pretrained on MIMIC-CXR, $\delta = 6\times10^{-6}$, clipping norm 1.5, 150 epochs, per-image guarantee. UKA-CXR (University Hospital Aachen, not public): 193,311 frontal chest X-rays from 45,016 patients; 153,502 training and 39,809 test images; 8 labels. Second dataset: 1,625 pancreatic-cancer CT scans, AUROC 99.28 at $\varepsilon = 8$ vs. 99.70 without DP; women had lower AUROC at every ε. nature

ε	0.29	0.54	1.06	2.04	4.71	7.89	no DP (∞)
Average AUROC (%)	83.13	84.00	84.98	85.80	86.93	87.36	89.71

Group	ε = 0.29	ε = 7.89	no DP
Male	82.66	87.12	89.46
Female	83.85	87.66	90.06
Age under 30	86.47	89.72	91.64
Age 80 and over	81.27	85.48	88.18

The statistical parity difference for under-30 patients grew from +4.00 without DP to +7.05 at ε = 0.29; the sex gap did not correlate with privacy level (Pearson r = 0.13). nature nih The age gap (under 30 vs. 80+) was 3.46 points without DP and 5.20 points at ε = 0.29. DP did not erase accuracy, but it widened the age gap. This is the measurement the team would make, but federated.

Securing Collaborative Medical AI by Using Differential Privacy: Domain Transfer for Classification of Chest Radiographs. Tayebi Arasteh et al., Radiology: Artificial Intelligence, 2024. nature Five chest X-ray datasets (VinDr-CXR, ChestX-ray14, CheXpert, UKA-CXR, PadChest), ε from 0.72 to 8.81. rsna External-validation performance "remained nearly unchanged across ε ≈ 1–10" (Mohammadi et al.), nature suggesting DP noise can help cross-site generalization. nih

The dermoscopy figure in the earlier brief ("69.1 → 43.0 at ε = 3") matches the review: the arXiv version of Mohammadi et al. (2506.00660) lists Guo, Yu et al. in Table 3 as "ε = 3/Accuracy = 43%" against a 69.1% baseline, and its text states that "accuracy declined from 69.1% to 43% at ε = 3."

1.4 Table: cost of DP at different ε (medical tasks)

Rows from Mohammadi et al. Table 3, except the UKA-CXR rows (Tayebi Arasteh et al. 2024). C = centralized; FL = federated.

Study	Modality	Setting	No DP	Strict ε (≈1 or below)	Moderate ε (≈3–10)
Tayebi Arasteh 2024	Chest X-ray (193,311 images)	C	AUC 89.7	ε = 0.29: 83.1; ε = 0.54: 84.0; ε = 1.06: 85.0	ε = 2.04: 85.8; ε = 4.71: 86.9; ε = 7.89: 87.4
Tayebi Arasteh 2024	Abdominal CT (1,625 scans)	C	AUC 99.7	ε = 0.5: 92.0	ε = 8.0: 99.3
Ziller et al.	Chest X-ray	C	AUC 96	ε = 0.5: 84	—
Riess et al.	Brain MRI	C	Acc 90.1	ε = 1: 81.6	ε = 10: 90.0
Guo et al.	Dermoscopy	C	Acc 69.1	—	ε = 3: 43; ε = 10: 61
Adnan et al.	Histopathology	FL	Acc 82	ε = 2.9: 81	ε = 10: 78
Mehmood et al.	Brain MRI	FL	Acc 88	—	ε = 5.6: 78
Odeyomi et al.	Chest X-ray	C	Acc 92	ε = 2: 88	ε = 8: 90
Chin-Cheong et al.	EHR (synthetic data)	C	AUC 80	ε = 1: 66	ε = 10: 68
Tayebi Arasteh (speech)	Pathological speech	C	Acc 99.1	ε = 0.9: 88.3	ε = 7.5: 95.3

Interpretation: pretrained chest X-ray and CT models lose 2–3 AUC points at ε ≈ 8 and 6–8 points at ε ≈ 0.5. Smaller, more varied datasets lose much more. nih Few federated rows exist and none include subgroups. Fed-Heart-Disease, with a few hundred patients per site, is in the "small and heterogeneous" group, so expect large losses at ε ⩽ 1.

1.5 Chart 1: AUROC vs. ε on UKA-CXR (centralized DP-SGD)

Source: Tayebi Arasteh et al. 2024. Axis 80–90 AUROC; one block = 0.5 point.

The curve rises quickly up to ε ≈ 2 and then flattens; the remaining 2.35-point gap at ε = 7.89 closes only when DP is removed.

1.6 Chart 2: local vs. federated performance in EXAM (20 hospitals)

Source: Dayan et al. 2021. Axis 0.60–1.00; one block = 0.025 AUC.

The larger benefit is on other hospitals' data (+0.253 AUC) than on the home hospital (+0.125 AUC). harvard Always report performance on every hospital's test set.

1.7 Attacks and audits

Do Gradient Inversion Attacks Make Federated Learning Unsafe? Ali Hatamizadeh, Hongxu Yin, ..., Holger R. Roth (NVIDIA), IEEE Transactions on Medical Imaging 42(7):2044–2056, 2023 (arXiv 2202.06924). Nine simulated clients classifying chest X-rays (COVID-19 vs. normal) with an ImageNet-pretrained ResNet-18; 1,360 training images in total; client sizes 1–512 images; batch sizes 1, 4, or 8; recorded with NVIDIA FLARE. researchr arxiv Findings:

- Earlier attacks fail when batch-norm statistics update during training or when the attacker lacks the current global model. arxiv
- Only the client training on 1 image with batch size 1 leaked an identifiable image (identifiability precision 1.0 up to noise level $\sigma_0 \approx 20$); all others stayed below 0.1 even at low noise. arxiv
- Larger datasets, batches, and more local iterations make attacks harder. arxiv

- Gaussian noise on updates (no formal ε) reduced leakage but also accuracy. arxiv
- Mitigations: hide the global model state (for example with HE), keep batch-norm statistics local (FedBN), use DP. arxiv
- Conclusion: "previous efforts likely exaggerated the risks of gradient inversion attacks in FL." arxiv

What actually works: gradient inversion is a real threat mainly for clients with very few records or a malicious server that alters the model. Membership inference is the more realistic risk for released models and is what DP directly bounds. So DP matters most for small sites (Fed-Heart-Disease's Switzerland site has 46 patients), and an audit should report membership inference advantage next to the formal ε.

How privacy is audited. Mohammadi et al. Table 6 lists 19 attack evaluations on DP medical models: mostly membership inference, plus gradient inversion, model inversion, reconstruction, and attribute disclosure. nih There is no standard protocol; most test one attack at one ε.

1.8 Fairness under DP and FL

What has been measured:

- Centralized DP, imaging: Tayebi Arasteh et al. found the age gap widened under strict DP while the sex gap did not change systematically; Kaess et al. (chest CT) examined race, age, and sex. nih nih
- Centralized DP, general ML: "Differential privacy has disparate impact on model accuracy" (cited in PUFFLE) arxiv showed DP-SGD hurts underrepresented groups more, because their larger gradients get clipped more. neurips
- Federated, no DP: a GitHub reproduction (JordanManoj/fair-fl-medical, not peer reviewed) reports FedAvg accuracy 0.748 on Fed-Heart-Disease, balanced accuracy 0.704 (FedAvg), 0.764 (federated normalization), 0.795 (pooled), with the sex parity difference widening −0.16, −0.24, −0.32. github If correct, better accuracy came with a larger sex gap. Treat as unverified.
- Federated with DP, large EHR: a 2026 Informatics in Medicine Unlocked study of 30-day readmission across 47 institutions reports formal DP and a maximum subgroup AUROC gap of 6.6% (uninsured vs. others). sciencedirect We saw only summary text; it does not appear to sweep ε.

Mitigations:

- Fairness across hospitals (not demographic groups), without DP: FairFed, q-FedAvg (upweights high-loss clients), Ditto (personalized FL).
- Fairness with DP in FL, non-medical: PUFFLE (arXiv 2407.15224); Xue and Yu (arXiv 2603.24392, March 2026), $\varepsilon \in \{0.75, 1, 2, 3, 4, 10\}$, 1–6 simulated sites, Adult Census and Law School data. arxiv

- Group-aware clipping, centralized: SoftAdaClip (arXiv 2510.01447, October 2025) on MIMIC-III text and GOSSIS-eICU reports subgroup gaps cut by up to 87% vs. DP-SGD. arxiv Seen only as a summary.
- Compliance-weighted noise: Parampottupadam et al. (arXiv 2505.22108, July 2026), 16 clients on PneumoniaMNIST and BreastMNIST, but no formal client-level DP and no demographic metric. arxiv

Status of Idea D (October 2026): our searches and a targeted search of January 2025–October 2026 found no paper combining (a) federated training across naturally partitioned medical sites, (b) DP at several ε, and (c) subgroup fairness by sex, age, or race. Each has at most two. The gap is open but is an obvious next step for the Tayebi Arasteh and Ziller groups, so the team should move quickly and post a preprint.

1.9 Aggregation methods on medical data

- FLamby: FedOpt variants best where FL helps; Cyclic worst; FedAvg competitive; FedProx and SCAFFOLD not consistently better. arxiv arxiv
- Zhou et al. 2025: no algorithm consistently best; many underperform on medical imaging. arxiv
- FedBN keeps batch-norm layers local; suited to scanner differences and reduces gradient inversion risk, but yields one model per site. researchgate arxiv
- FeTS 2025: adaptive aggregation helped on average; worst-site failures remained. lih nature
- MOON, Ditto, q-FedAvg: we found no head-to-head medical numbers. MobenFL ("Benchmark Evaluation of Federated Learning on Multi-organ Images", arXiv 2607.08219; 22 datasets, 12 organs, 20 algorithms) is the place to look. arxiv

Conclusion: baselines should be local, pooled, FedAvg, FedProx, FedAdam, and one personalization method (FedBN for imaging, or FedAvg plus local fine-tuning for tabular). A new aggregation rule is unlikely to win clearly.

1.10 DP methods compared

Method	What is protected	Where noise is added	Fits 4 hospitals?	Fits 200 hospitals?	Notes
DP-SGD inside each client (sample-level)	Each patient record	Each client, every local step	Yes	Yes	Opacus; compose over all rounds with an RDP accountant
DP-FedAvg, central DP (client-level)	Whether a whole hospital took part	Server, after clipping each update	No: noise must hide a whole hospital among 4	Yes, with many clients sampled per round	Flower: DifferentialPrivacyClientSideFixedClipping or ServerSide variants flower
Adaptive clipping	Same	Same, with clipping norm learned from a quantile of update norms	Same	Same	Flower: DifferentialPrivacyClientSideAdaptiveClipping with adaptiveclipping_mod flower
Local DP on updates	Update even from the server	Each client before sending	Possible, very noisy	Possible, very noisy	Flower: LocalDpMod flower
Secure aggregation	Individual updates	No noise	Yes	Yes	No ε by itself; Flower has SecAgg and SecAgg+
Homomorphic encryption	Updates during aggregation	No noise	Yes	Yes, with large compute cost	NVIDIA FLARE supports HE; MedHE preprint reports 54% overhead arxiv

Why client-level DP fails with 4 hospitals: the model must look almost the same whether or not one entire hospital took part. With 4 hospitals, one hospital is 25% of the signal, so the noise must be about as large as one hospital's clipped update every round. Flower's documentation recommends a noise multiplier of 1.0 or higher for strong privacy; flower with 4 clients and full participation, this noise is comparable to the averaged update, and there is no amplification from sampling. This is the main correction to the earlier Idea D: on FLamby's 2–6 client datasets use sample-level DP; reserve client-level DP for eICU (208 hospitals).

MedHE (arXiv 2511.09043, November 2025) combines top-k sparsification with CKKS homomorphic encryption and reports communication for 5 clients falling from 1,277 MB to 30.5 MB (42×) with 54% extra training time. arxiv It is a single-author preprint that also claims "HIPAA compliance" and scaling to "100+ institutions"; do not rely on it without reproduction. arxiv

1.11 Communication cost

Only 19 of 107 healthcare FL studies measured communication (Li et al. 2025). arxiv Bytes per round = parameters × 4 bytes (32-bit floats) × clients × 2 (upload and download). Fed-Heart-Disease logistic regression with 13 inputs has 14 parameters, 56 bytes per client per direction, so communication does not matter there. arxiv EfficientNet-B0 on Fed-ISIC2019 (about 5.3 million parameters, a standard figure) is about 21 MB per client per direction per round. Flower's message_size_mod and arrays_size_mod log message sizes. flower flower

1.12 Foundation models and federated fine-tuning under DP

LoRA (low-rank adaptation) trains small added matrices instead of all weights, so clients send far fewer numbers. 2024–2026 work includes DP-LoRA (federated, medical text tasks, ACM), DP-FedLoRA (arXiv 2509.09097), FedSVD (arXiv 2505.12805), SDFLoRA (arXiv 2601.11219), WinFLoRA (arXiv 2602.01126), Fed-HeLLo (2025), and FediLoRA (IJCAI 2026, with a medical dataset). arxiv +4 Most use general text benchmarks and simulated splits. The area is crowded and compute-heavy, so the old Idea C (federated LoRA rank allocation) does not fit a first-year team with laptops. A lighter option is a frozen pretrained medical encoder with only a small head trained by FL plus DP, supported by the DP review's finding that pretraining is often necessary and the FL review's finding that only 5 of 107 studies used pretrained models. nih arxiv

1.13 Tools

Tool	What it does	Client-level DP	Sample-level DP	Notes
Flower (flwr, v1.39 docs)	FL framework: simulation and deployment	Yes: ClientSide/ServerSide Fixed and Adaptive Clipping wrappers; fixedclipping_mod and adaptiveclipping_mod	Through Opacus inside the ClientApp (official example)	Also LocalDpMod, SecAgg/SecAgg+, size mods; v1.39 API lists PrivacyAccountant and RdpAccountant under flwr.serverapp flower
Opacus (PyTorch)	DP-SGD with RDP accountant	Not directly	Yes	Replace BatchNorm with GroupNorm
TensorFlow Privacy	DP-SGD for TensorFlow/Keras	Not directly	Yes	Only if using TensorFlow
Google dp-accounting	Stand-alone accountants (RDP, privacy loss distribution)	Can account DP-FedAvg	Yes	Use to double-check ε
FLamby	7 naturally split datasets, baselines, FL strategies	Small example	Small example	Plugs into Flower via PyTorch DataLoaders
NVIDIA FLARE	Production FL framework (used in EXAM)	DP filters	Yes	Supports HE; heavier to set up

Recommendation: Flower plus FLamby, Opacus for sample-level DP, and check every ε with a second accountant.

1.14 Datasets

Dataset	Modality	Task	Size	Natural sites	Access
Fed-Heart-Disease (FLamby)	Tabular	Binary (heart disease)	740 patients; 303 / 261 / 46 / 130; 13 features	4 (Cleveland, Hungary, Switzerland, Long Beach VA)	Open (UCI)
Fed-TCGA-BRCA (FLamby)	Tabular clinical	Survival (C-index), Cox model	1,088 patients (51–311 per region); 39 features	6	Open
Fed-ISIC2019 (FLamby)	Dermoscopy	8-class (balanced accuracy)	23,247 images; 12,413 / 3,954 / 3,363 / 2,259 / 819 / 439; 9 GB	6	Open
Fed-Camelyon16 (FLamby)	Pathology slides	Tumor (AUC)	399 slides (239 / 150); 50 GB features	2	Open (large)
Fed-IXI (FLamby)	Brain T1 MRI	Brain mask (Dice)	566 scans (311 / 181 / 74); 444 MB	3	Open
Fed-KITS2019 (FLamby)	Abdominal CT	Kidney/tumor (Dice)	96 cases (12–30 per site); 54 GB	6	Open
Fed-LIDC-IDRI (FLamby)	Chest CT	Lung nodule (Dice)	1,018 scans (670 / 205 / 69 / 74); 115 GB	4 (scanner maker)	Open via TCIA
eICU-CRD v2.0	ICU EHR	Mortality, stay length, readmission	200,859 unit stays, 139,367 patients, 335 ICUs, 2014–2015	208 hospitals	PhysioNet credentialed (CITI training plus data use agreement) nature

Dataset	Modality	Task	Size	Natural sites	Access
MIMIC-IV (v3.1)	Hospital/ICU EHR	Many	76,540 ICU admissions, 2008–2019	1	PhysioNet credentialed researchgate
MIMIC-CXR	Chest X-ray plus reports	Multi-label	377,110 chest X-rays from 227,827 imaging studies, Beth Israel Deaconess, 2011–2016 (Johnson et al., MIMIC-CXR-JPG)	1	PhysioNet credentialed
CheXpert	Chest X-ray	Multi-label	224,316 chest radiographs of 65,240 patients (Irvin et al., AAAI 2019)	1 (Stanford)	Registration and agreement
MedMNIST v2	Small 2D/3D images	18 tasks	708,069 2D and 10,214 3D images from 12 2D and 6 3D datasets (Yang et al., Scientific Data 2023)	None	Open
FeTS / BraTS	Brain MRI	Tumor (Dice)	1,473 cases with institution labels (FeTS 2024)	Dozens	Registration (Synapse)
UKA-CXR	Chest X-ray	8 labels	193,311 images, 45,016 patients	1	Not public

MIMIC-IV, MIMIC-CXR, and CheXpert are single-site, so "hospitals" made from them are artificial splits. eICU is the best large, naturally partitioned EHR benchmark emergentmind and has sex, age, and ethnicity fields. Fed-ISIC2019 is the best imaging choice for laptops plus PACE. MIMIC-CXR, CheXpert, and MedMNIST sizes are taken from their original dataset papers (Johnson et al.; Irvin et al.; Yang et al.).

1.15 Venues and deadlines (status as of 7 October 2026)

Venue	Most recent verified dates	2027 status	Fit
DeCaF (MICCAI workshop)	DeCaF 2026: deadline extended from 1 July to 24 July 2026; notification 12 August; held 1 October 2026 in Strasbourg; 8-page LNCS, double-blind, OpenReview github	Not announced; expect a July 2027 deadline	Topics include "privacy-preserving techniques (e.g., secure aggregation, differential privacy)" and fairness in heterogeneous FL; github best if imaging is included
MICCAI 2027 main	One tracker estimates late February 2027 (not official) trybibby	Not announced	Too early
ML4H (AHLI)	ML4H 2026: deadline 10 September 2026; decisions 22 October; held 6–7 December 2026 in Sydney; Proceedings (8 pages) and Findings (4 pages) ahli ahli	Not announced; expect early September 2027	Good fit for tabular/EHR audit
CHIL (AHLI)	CHIL 2026: deadline 4 February 2026; held 28–30 June 2026 in Seattle; 8–10 pages ahli x	Not announced; expect early February 2027	Stretch; needs results by January 2027
NeurIPS/ICML/ICLR workshops, SaTML, PETS	Not verified	Unknown	Check in spring 2027

Correction: DeCaF is held at MICCAI in late September or early October; July is the submission deadline, not the event. github

Document 1: Possible research directions for the team

Summary table

#	Direction	Gap it fills (source)	Data	Compute	Risk	Venue	Verdict
1	Privacy–fairness audit of federated hospital models across ε	No federated multi-ε subgroup audit found; Mohammadi et al. 2026 call for it	Fed-Heart-Disease → eICU (208 hospitals); optional Fed-ISIC2019	Laptop for tabular; PACE GPU for ISIC	Medium	ML4H 2027, DeCaF 2027, CHIL 2027 stretch	Pick this
2	Honest evaluation harness (per-site tests, worst site, local/fed/pooled, bytes, ε)	84% lack per-client splits; 19/107 measure communication (Li et al. 2025)	FLamby tabular, Fed-ISIC2019	Laptop	Low	ML4H Findings, DeCaF	Build first as the base of #1
3	Worst-hospital performance under DP	FeTS 2025: averages hide site failures	Fed-ISIC2019, eICU	PACE GPU	Medium–high	DeCaF 2027	Fold into #1
4	Empirical privacy audit (MIA vs. formal ε)	Attacks rarely tested at multiple ε; Hatamizadeh 2023 says risks may be overstated	Fed-Heart-Disease, eICU	Laptop/PACE	Medium	Privacy workshops (unverified)	Spring add-on
5	Frozen encoder + federated DP head	Pretraining key under DP; 5/107 FL studies used pretrained models	Fed-ISIC2019, MIMIC-CXR	PACE GPU	Medium	DeCaF 2027	Backup if imaging is chosen

Dropped: federated LoRA rank allocation (old Idea C), because of many 2025–2026 papers and large GPU needs.

Direction 1 (recommended): Privacy–fairness audit

Gap. Only five of 74 DP medical studies audit fairness, most at one ε, and multi-ε ones are centralized (Mohammadi et al. 2026). arxiv +2 No 2025–2026 federated study found sweeps ε with sex, age, or race subgroups on naturally partitioned medical data.

What to build and measure.

1. Local, pooled, and FedAvg/FedProx/FedAdam models, each with and without DP.
2. DP setting A (sample-level, all datasets): Opacus DP-SGD in each hospital; $\varepsilon \in \{0.5, 1, 3, 8, \infty\}$; $\delta \leq 1/(\text{training records at that hospital})$; RDP accountant composed over all rounds.
3. DP setting B (client-level, eICU only): Flower's ClientSideFixedClipping and ClientSideAdaptiveClipping with many hospitals sampled per round.
4. For each ε: overall AUC, per-hospital AUC, worst-hospital AUC, AUC by sex and age group (for example under 55 vs. 55+), subgroup gaps, statistical parity and equal opportunity differences, bootstrap 95% confidence intervals.
5. One mitigation: subgroup-aware clipping or reweighting, compared with Flower's adaptive clipping.

Questions. Does the subgroup gap grow as ε shrinks in FL, as age did centrally? Does FL widen or narrow the gap vs. local and pooled? Does group-aware clipping reduce the gap at fixed ε?

Datasets. Fed-Heart-Disease (740 patients, 4 sites; has sex and age) for development; eICU-CRD (200,859 stays, 208 hospitals; sex, age, ethnicity) for main results; nature optionally Fed-ISIC2019 (23,247 images, 6 sites; confirm age and sex metadata in the FLamby loader).

Compute. Fed-Heart-Disease: seconds per run; 5 ε × 3 strategies × 5 seeds = 75 runs, under an hour on a laptop. eICU with a small MLP: minutes per run on CPU; PACE for sweeps. Fed-ISIC2019 EfficientNet: GPU hours per run on PACE.

Risks. (a) Tiny test sets (Switzerland site: 46 patients total) give wide intervals, hence eICU. (b) Client-level DP with few clients will ruin the model; report it as a finding. (c) Another group may publish first; post a preprint once eICU results exist. (d) eICU credentialing takes time; start now.

Venue. ML4H 2027 for EHR results; DeCaF 2027 if Fed-ISIC2019 is included; CHIL 2027 only if eICU results are solid by January.

Direction 2: Honest evaluation harness

Gap. 84% of healthcare FL studies lack per-client splits, 19 of 107 measure communication, 26 of 107 address any FL-specific issue (Li et al. 2025); arxiv FLamby lacks privacy. arxiv **Build** a harness reporting per-hospital and worst-hospital metrics, local/federated/pooled comparisons, bytes, wall-clock time, and (ε, δ) with accountant name. It is the base for Direction 1. **Risk:** low; modest novelty alone.

Direction 3: Worst-hospital performance under DP

Gap. FeTS 2025 showed good averages hid site failures; lih DP hurts small sites most. **Build:** compare FedAvg, q-FedAvg, and FedBN or local fine-tuning on worst-hospital AUC at each ε. **Risk:** medium–high, because personalization changes which model is evaluated.

Direction 4: Empirical privacy audit

Gap. Attacks are usually tested at one ε (Mohammadi et al. Table 6). **Build:** a loss-threshold membership inference attack per hospital and ε, reporting attack AUC next to formal ε, separately for small and large sites. **Risk:** medium.

Direction 5: Frozen encoder plus federated DP head

Gap. Pretraining is "often necessary" under DP, nih but only 5 of 107 FL studies used pretrained models. arxiv **Build:** extract Fed-ISIC2019 features once with a frozen encoder; train a linear or small head with FedAvg plus DP; sweep ε. **Risk:** medium; depends on the encoder.

Recommendation

Pick Direction 1, built on Direction 2's harness, with worst-hospital results from Direction 3 as a secondary metric. The gap is confirmed by the largest review and a fresh 2025–2026 search; it needs little compute; it reuses Orlando's code; it yields results even if the mitigation fails, because the audit is the contribution; and it matches the course goal to "evaluate behavior under privacy requirements … with clear experiments and metrics." The one change from the earlier brief: sample-level DP on the 4-hospital FLamby datasets, client-level DP only on eICU.

Document 2: What Orlando should do

Step 1. Understand each part of the FedAvg loop

1. **Client selection (server, Strategy):** which clients train this round (fraction_fit). With 4 FLamby hospitals, select all 4 (FLamby uses full participation). Check: print client IDs each round.
2. **Broadcast:** server sends current global parameters.
3. **Local training (ClientApp fit):** client loads parameters, runs a fixed number of local steps (FLamby: 100 batches), returns new parameters and its number of examples. arxiv Check: local loss drops within the round.
4. **Aggregation (server):** weighted average by number of examples. Check: with one client, the global model equals that client's model.
5. **Evaluation:** each client evaluates the global model on its own test set and returns metric and test size. Check: one number per hospital, not just an average.

Write a one-page notebook note explaining these five parts with the file and function where each happens.

Step 2. Add to the harness, in this order

1. **Per-client test sets:** FLamby's fixed splits (Fed-Heart-Disease 199/104, 172/89, 30/16, 85/45); accuracy and AUC per center. github
2. **Local, federated, pooled:** train all three and evaluate each on every center's test set (a 3 × 4 table).
3. **Worst site and uniform average** (FLamby's convention), not only the sample-weighted average.
4. **5 seeds** with mean and standard deviation; bootstrap intervals for subgroups.
5. **Bytes counter** via message_size_mod or parameters × 4 bytes. flower
6. **FedProx and FedAdam** (built into Flower).
7. **Sample-level DP:** Opacus PrivacyEngine on each client's optimizer; compose privacy over all rounds; report per-hospital ε with $\delta \leq 1/n$; check with a second accountant.
8. **ε sweep** $\{0.5, 1, 3, 8, \infty\}$.
9. **Subgroup metrics** by sex and age: AUC, accuracy, true and false positive rates, statistical parity and equal opportunity differences.
10. **Client-level DP comparison:** Flower's DifferentialPrivacyClientSideFixedClipping with fixedclipping_mod flower on Fed-Heart-Disease; flower record the accuracy drop as evidence for using sample-level DP with 4 hospitals.

Step 3. Read in this order

1. McMahan et al., AISTATS 2017 (FedAvg), Sections 1–3.
2. FLamby, NeurIPS 2022, Sections 3.2, 3.3, 4, Appendix L.1.
3. Li et al., Medical Image Analysis 2025, Section 4 and 5.2.2.
4. Flower docs: "Differential Privacy" explanation and "Use Differential Privacy" how-to. flower
5. Abadi et al., CCS 2016: DP-SGD and the moments accountant.
6. Mohammadi et al., npj Digital Medicine 2026: background, Table 3, fairness section.
7. Tayebi Arasteh, Ziller et al., Communications Medicine 2024: the ε table and subgroup analysis (template for your figures).
8. Hatamizadeh et al., IEEE TMI 2023: Results and Table II.

Step 4. Datasets to run next

- Now: Fed-Heart-Disease.
- Next 2 weeks: Fed-TCGA-BRCA (1,088 patients, 6 sites, C-index) to test a second dataset and metric. arxiv
- This week: start eICU credentialing on PhysioNet (CITI "Data or Specimens Only Research" course plus data use agreement); approval can take days to weeks. scite

- November or spring: eICU grouped by hospital (or US region if hospitals are too small). arxiv
- Spring with PACE: Fed-ISIC2019.

Step 5. Notebook and design document

Weekly notebook entry: date; goal; runs (dataset, strategy, ε, seeds, commit hash); results table; one plot; problems and fixes; papers read with two-sentence summaries; time spent; your contribution vs. teammates'; next week's plan.

Design document (before major implementation; mid-term documentation due the week of 5 or 12 October):

1. Problem: hospitals cannot share data; FL plus DP protects patients but may harm some groups more.
2. Requirements: per-patient privacy at stated (ε, δ); per-hospital and per-subgroup reporting; fixed seeds.
3. Threat model: honest-but-curious server; attacker sees updates or the final model.
4. Approach: Flower + FLamby + Opacus; FedAvg, FedProx, FedAdam; local and pooled baselines; sample-level DP; ε sweep.
5. Evaluation: AUC, accuracy, worst-site AUC, subgroup gaps, bytes, ε; 5 seeds; bootstrap intervals.
6. Risks and fallbacks: small test sets → eICU; client-level DP failure → report as a finding.
7. Timeline (below).

Step 6. Week-by-week plan (7 October – 8 December 2026)

Week	Dates	Tasks	Deliverable
1	Oct 5–11	Design document; five-part FedAvg note; start eICU credentialing	Mid-term design document (if due)
2	Oct 12–18	Per-client tests; local/federated/pooled table; worst site; 5 seeds	Mid-term documentation (if due); first results table
3	Oct 19–25	Bytes counter; FedProx, FedAdam; Fed-TCGA-BRCA; read papers 1–4	Baseline table, two datasets
4	Oct 26–Nov 1	Opacus sample-level DP; second-accountant check; read papers 5–6	First accuracy vs. ε curve
5	Nov 2–8	ε sweep × 3 strategies × 5 seeds; sex and age metrics with bootstrap; read paper 7	Subgroup gap vs. ε plot
6	Nov 9–15	Client-level DP (fixed and adaptive clipping) vs. sample-level; prepare 20-minute tutorial on DP-SGD and DP-FedAvg	Tutorial slides
7	Nov 16–22	One mitigation (group-aware clipping or reweighting); load eICU if approved; read paper 8	Mitigation result
8	Nov 23–29	Thanksgiving: clean code, freeze results, write final report	Draft final report
9	Nov 30–Dec 8	Final presentation; notebook check	Final presentation and documentation

Step 7. Prepare for spring 2027

1. Full audit on eICU (sample- and client-level DP; sex, age, ethnicity; worst hospital).
2. Fed-ISIC2019 on PACE if aiming at DeCaF 2027.
3. Membership inference audit (Direction 4).
4. Decide the venue by mid-January: CHIL 2027 if eICU results are complete (expected early February, not announced); otherwise DeCaF 2027 (expected July) and ML4H 2027 (expected September).
5. Post an arXiv preprint once eICU results are stable.

Caveats

- FLamby reports results only as figures; take exact numbers from its GitHub benchmark files before quoting them.
- The UKA-CXR fairness numbers are centralized, single-hospital, record-level DP results, not federated.
- The fair-fl-medical numbers come from an unreviewed GitHub repository; MedHE's figures from an unreviewed single-author preprint.
- The 47-institution readmission study and SoftAdaClip were seen only as summaries.
- "No federated multi-ε subgroup audit exists" is strong evidence from our searches, not proof.
- 2027 deadlines for DeCaF, CHIL, ML4H, and MICCAI are estimates from past years; NeurIPS/ICML/ICLR workshop, SaTML, and PETS deadlines were not verified.
- MIMIC-CXR, CheXpert, and MedMNIST sizes are taken from their original dataset papers (Johnson et al.; Irvin et al.; Yang et al.); the MIMIC-CXR-JPG description gives image and study counts but no patient count.